

Intelligent Ambulance with Automatic Traffic Control

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Abstract — The existing predictable traffic signal timing methods does not result in an effective control of congestion. An intelligent traffic control system for congestion control, gives way for the emergency vehicle. For each individual vehicle the tag has been placed at a spot for making it impossible to eliminate or destroy. The tag which is attached to vehicles has been read by RFID readers with Arduino. The reader counts the number of vehicles passing it for a specific duration. By determining the congestion, the green light is turned on for the path with highest number of vehicles. The arrival of the emergency vehicle is detected using XBEE and the green light is turned on for that particular path.

Keywords — **Index Terms**— XBEE, Arduino mega, Personal Computer, Ambulance Vehicle, Traffic Junction

I. INTRODUCTION

Now a day's many people preferred to travel in a vehicle to meet their day-to-day needs. Nobody has enough time to waste their time by waiting at traffic junctions. But gridlocked expressways take the mobile out of the automobile. Traffic congestion plays a major role across India. During peak hours there is heavy traffic. It origins from the widespread desire of people to track certain goals that predictably overload existing roads and transit systems daily. Hence an intelligent traffic management system is mandatory in the prevailing situation. Being cost-effective wireless networks are widely used in road transport in recent years. We developed a system using Radio frequency identification (RFID) Technology, Global system for mobile communication (GSM) module, XBee and advanced Arduino kit to achieve the desired results. Here RFID technology is used for congestion control. It is a wireless technology that uses radiofrequency EM wave for transmission and reception of frequency. A network-connected device with an antenna is an RFID reader. It sends power, data and commands to the tags. Performing many tasks including simple continuous inventorying, filtering and writing is done by the reader which communicates with tags. RFID tags are classified into two types- active tag and passive tag. Battery-powered tags are active tags and the tags do not have onboard batteries are passive tags. It consists of various ranges and those ranges depend on the transmit power, antenna used, frequency, tag orientations, surroundings, etc. A high-level communication protocol is used for the emergency case is XBee. XBee is simpler compared to other WPAN networks like Bluetooth, Wi-Fi, etc.

II. LITERATURE SURVEY

For emergency vehicles, ZIGBEE has been used, but it has only a range of 30 meters. Detecting an emergency vehicle before 30 meters makes it impossible to take instant action.

The green light is turned on for the side with maximum congestion. In case, if the congestion remains to be high on that specific side, creating delay for the other three sides. Once the stolen vehicle comes into the range of RFID readers, the red light turns and stays red until the vehicle stays in the reader range, creating a mess. A vehicle passing through a green signal will continue to receive green signals as it travels through the road in a 'green wave' setup. The disadvantage of the green waves is when the wave is disturbed the density of vehicles in a green wave grows in size. Every time when the ambulance drivers have to update the priority of the vehicle, location and a total number of junctions has to be passed by using the GSM module. Modern society aims for automation, but for this situation, updating has to be made manually. A camera is used to detect the congestion at the intersection. Using cameras don't provide accurate results in bad weather condition. In all the above models, the algorithm for either of the congestion control or paving way for an emergency vehicle or stolen vehicle detection has not been explained.

III. PROBLEM DESCRIPTION

Now a days the green light dynamically changes in a rotational fashion with specific time delay. In case of absence of any vehicle in a particular lane, the green has to be still turned on if it is its turn and the vehicles in the other lane have to wait. In case of arrival of emergency vehicle, people will pave way by their own sense or a traffic police has to be present to control the traffic at the junction.

IV. PROPOSED MODEL

This intelligent traffic control system controls the congestion, paves way for emergency vehicle and detects the stolen vehicle at four-lane. Four RFID readers are placed in each lane, at a distance of 100 meters from the traffic junction. The RFID tags adhere to every individual vehicle all over the country. When a vehicle owning an RFID tag crosses the reader it acts as an access point and communicates with tags within its field range. The antenna in the reader captures data from tags and sends it to the computer for processing. Whenever the reader reads a tag, the value of the counter will be increased by one. Hence the total number of vehicles in a specific lane can be detected with counter's value. Based on the congestion, the timing for the green signal is allotted. The particular lane is ignored when there is no vehicle in the lane. When the emergency vehicle is taken for the purpose of an emergency, the driver must turn on the XBEE transmitter. The XBEE transmitter in-vehicle comes into range of XBEE receiver when the emergency vehicle approaches the traffic junction. In the junction, an interrupt is sent to

PC to check most recently read UID by all readers, with an emergency vehicle database stored in PC. If any read UID matches with any UID in the database an alarm is rung as a warning and the green light is turned on to the side with the emergency vehicle in 5 seconds. The owner will inform traffic control room when his vehicle is stolen. The UID number of the stolen vehicles will be identified and broadcasted to all the traffic junctions. GSM at traffic junction will receive the UID number and compares it with, UID numbers read by all the RFID readers. If any of the numbers match with the received number, a text message is sent to the traffic control room with timing details.

V. BLOCK DIAGRAM

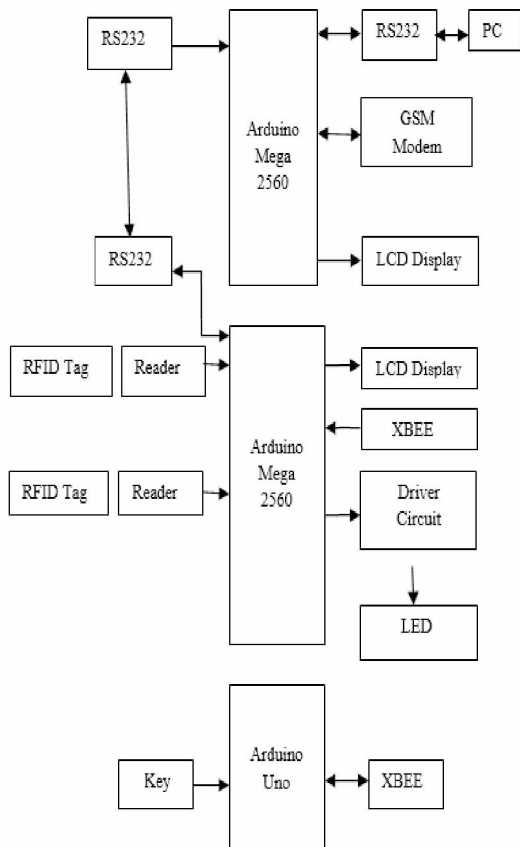


Fig. 1. Block diagram for signal control, stolen vehicle detection, emergency vehicle clearance

a. RFID TAG

Every RFID (Radio Frequency Identification) tag has UID (Unique Identification) number, chips, some memory and an antenna. The utilization of power is not required in passive tags and they are determined by an antenna that enables the tag to accept electromagnetic waves from a reader. Active RFID tags contain on board power source with greater memory and read at larger ranges. Because of its inbuilt power source, it has ability to act autonomously (without an external activation). The power runs the microchip's circuits and broadcasts a signal to the reader. The tag which falls in two frequency ranges- 433 MHZ and 915 MHZ is known as active tags. These tags can withstand high temperature and extreme moisture.



Fig. 2. Radio frequency identification tag

b. RFID READER

A Radio Frequency Identification reader (RFID) is a physical device which is used to collect the details of vehicle information from a RFID tag. The reader consists of a RF module where the sending and receiving of radio frequency signals is performed. Oscillators generate a carrier signal and modulate the RF signal with the carrier wave, to reinforce it. The receiver has a demodulator to extract the original data being transmitted and an amplifier to empower the received signal. RFID reader has three types; they are fixed, mobile and handheld units. The type of RFID reader is used based on the environment and application.



Fig. 3. RFID reader

c. XBEE

For a high level communication protocols an IEEE 802.15.4 XBEE is used. An XBEE embedded module is a tri combo: a form factor, a host interface, and a group of different protocols. It is designed for variety of purposes and low latency. Compare to other Wireless Personal Area Networks (WPANs), such as Wi-Fi or Bluetooth it is simple and less expensive. With the minimum number of

connections- power, ground, data in and data out the XBEE is used. Based on the type of XBEE, its range varies from meters to kilometers. XBEE offers quick and easy way to add wireless communication to any system.

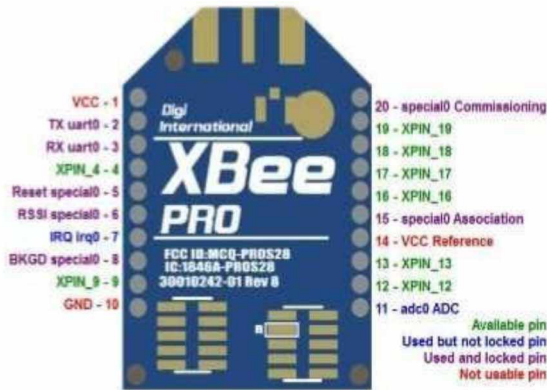


Fig. 4. XBEE

VI. WORKING MODEL

The entire algorithm of this paper is divided into three categories.

a. AUTOMATIC SIGNAL CONTROL SYSTEM

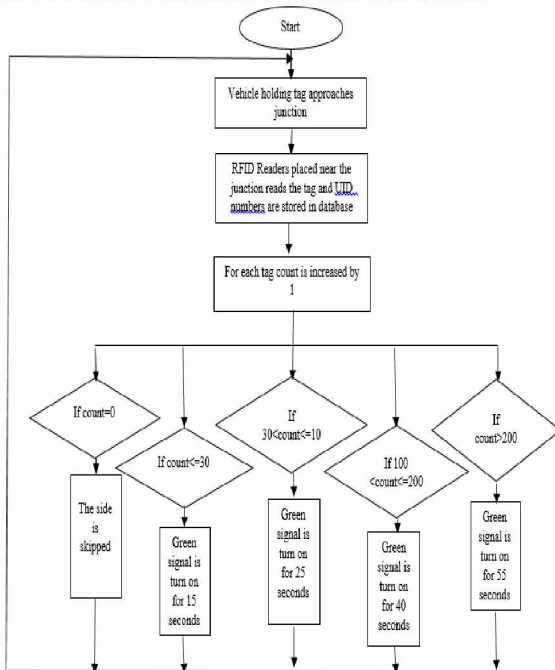


Fig. 5. Flowchart for congestion control

In this module, for prototype passive RFID tag and short range RFID reader have been used. The UID number will be transmitted to the reader, when the vehicle arrives into the range of the reader. The data is sent to the PC for storage. The Arduino controlled value is increased by one. Based on the congestion, the timing for the green signal is varied. XBee transmitter is placed in all the emergency vehicles and XBee receivers are placed in all the junctions. The emergency vehicle database is priority stored in all the personal computer at all junctions.

b. SIGNAL CLEARANCE FOR EMERGENCY VEHICLE

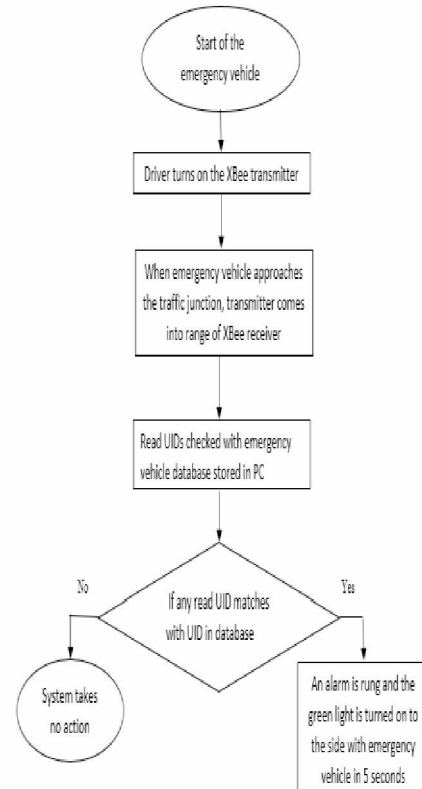


Fig. 6. Flowchart of signal clearance for emergency vehicle

XBee transmitter comes into the range of receiver, when the emergency vehicle arrives towards the junction. Instantly, an interrupt is sent to personal computer to check all the recently stored UID numbers. If any of the number matches with the number in the database, then an alarm is turned on as a warning to the side with green signal and the light changes from green to yellow. In 5 seconds, the green light is turned on for the side with emergency vehicle.

VII. RESULT

The program was successfully burned on the Arduino using USB programmer. On the basis of congestion, the timing for the green light changed and in case of absence of vehicle the particular side is skipped. When an emergency vehicle approaches the junction, it is successfully detected by the system as an emergency vehicle and necessary action is taken. On the other hand, if any stolen vehicle is detected, a message is sent to traffic control room. The system is effective, but in some of the cases to identify the vehicle the image processing based system is used, which has a major drawback during the bad climate environments. Bad weather may be due to heavy rain, fog, dust or storm [8-17]. This system is not affected by any of these bad weather conditions. Our system can work in any weather condition, so it has the capability to provide 24x7x365 surveillance without any disruption. The traditional system doesn't provide 24x7x365 surveillance. This system is very helpful in building a smart city. There won't be any issues related to traffic management if the city is well equipped. Moreover it will make cities safer in context of detection of stolen vehicles.

VIII. CONCLUSION

Realizable and relevant traffic system is proposed in this paper. Signal clearance is automatized based on the traffic congestion in the four lanes. XBEE, GSM SIM900 and RFID technologies are used in the proposed system. This system is fully automated, thus reducing man power. In case of arrival of emergency vehicle an alarm is rung as a warning and the green light is turned on to the side with emergency vehicle in 5 seconds. It avoids gridlock, which is the prevailing situation. At the four lanes, the stolen vehicle is detected by using GSM. The control room gets the information about the stolen vehicle, the message is broadcast to all the junctions and the vehicle is tracked.

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